

Thermal Comfort in the Historic Core: A Comparative Network Analysis of Delft and Xi'an

Group F

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03	Results
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Research
Plan

01

City Pair & Justification

Selected Cities

Delft, Netherlands & Xi'an, China

Selection Justification

- **Morphology**

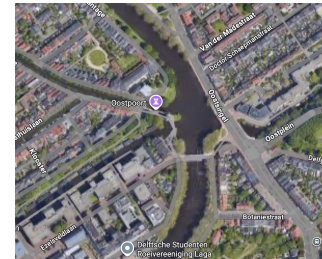
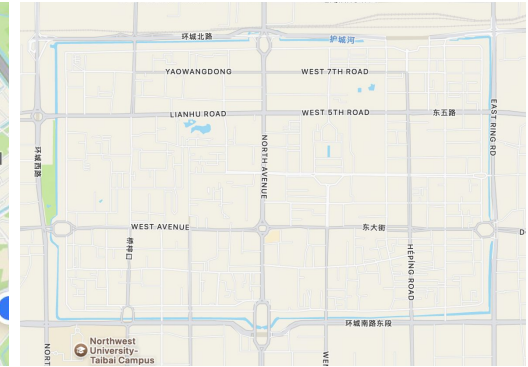
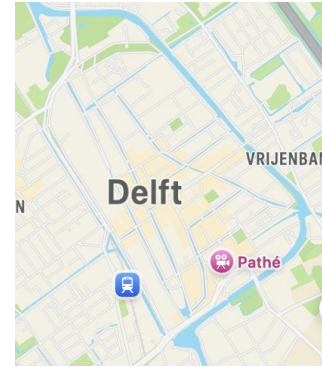
City	Delft	Xi'an
Contrast	Organic, compact, water-based medieval structure	Massive, rigid, grid-based walled city
Similarity	Historic centers in the core of urban area are separated from later urban expansion by a clear morphological & water-based boundary	

- **Shared Challenges**

Extreme summer heat stress in the dense historic centers

- **Scale Comparison**

"compact" city centre VS "mega-grid" city centre



Research Topic

Theme

Urban Climate (Heat)

Research Question

How does thermal comfort vary across pedestrian street segments within the historic city center?



Which pedestrian streets are structurally important within the urban network?



How can **thermally stressful conditions** along **important pedestrian streets** be **identified and optimized** in historic city centers?



Which pedestrian street segments should be prioritized for heat mitigation interventions based on thermal stress and network importance?

Research
Methodology

02

Data Collection

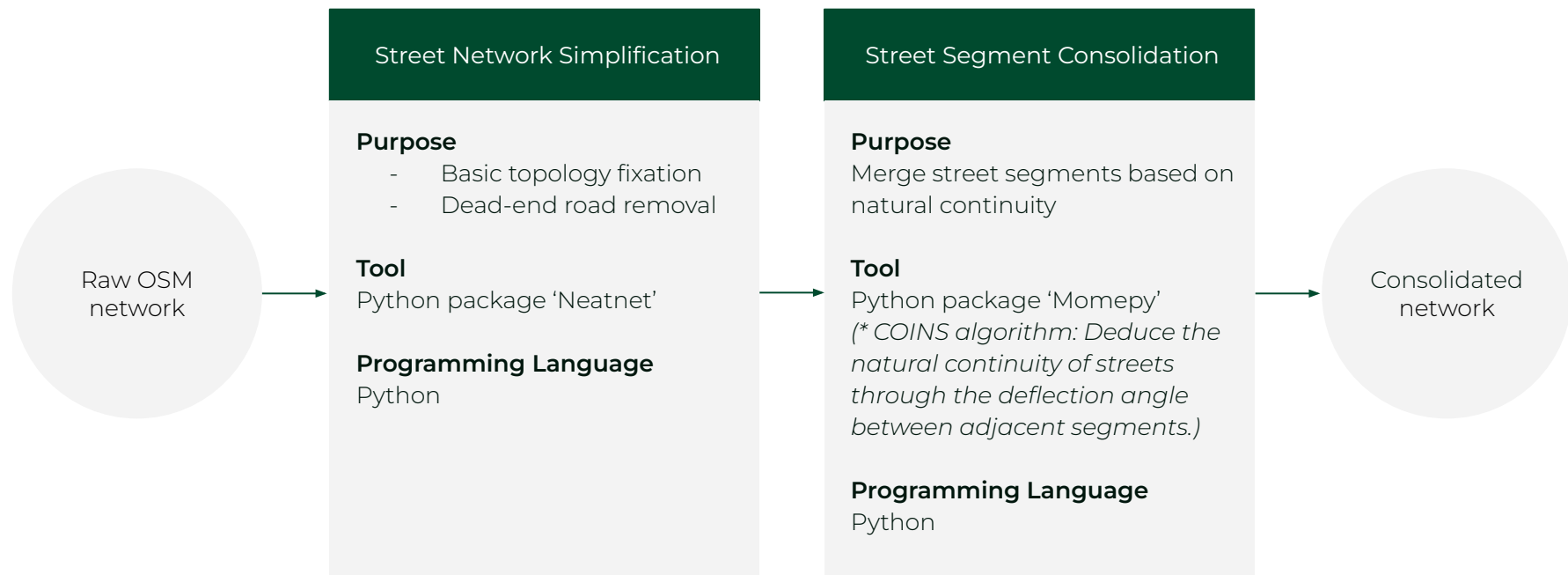
Dataset	Network	Land Surface Temperature	Green Space	Blue Space
Resource	Delft: PDOK Xian: OSM	Landsat 8-9 Level 2	OSM	OSM
Resolution	-	30m X 30m	-	-
Spatial Extent	City center + 500m buffer	City center + 500m buffer	City center + 500m buffer	City center + 500m buffer
Temporal Extent	-	Jul 2025	-	-

Methodology

01 Preprocess

02 Variable Calculation

03 Clustering



Methodology

01 Preprocess

02 Variable Calculation

03 Clustering

Which streets are structurally important within the network?

Popularity

How does thermal comfort vary across street segments?

Land Surface Temperature

What kinds of heat mitigation intervention are required?

Distance to Blue Space

Distance to Green Space

Methodology

01 Preprocess

02 Variable Calculation

03 Clustering

Popularity

Method

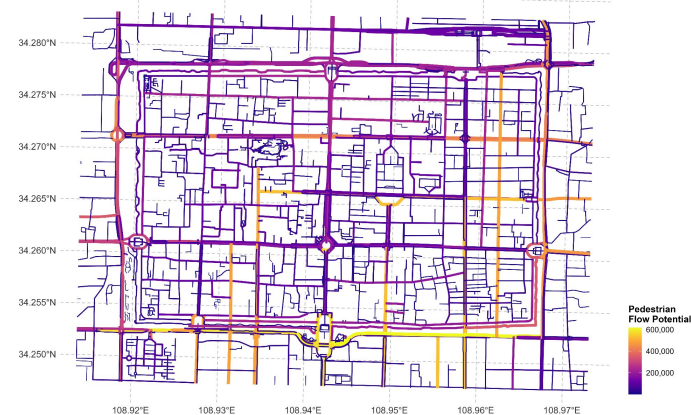
Angular Choice

Method Description

Calculate the number of times each street segment falls on the shortest path between all pairs of segments within a selected distance



Delft



Xian

Methodology

01 Preprocess

02 Variable Calculation

03 Clustering

Land Surface
Temperature

Method

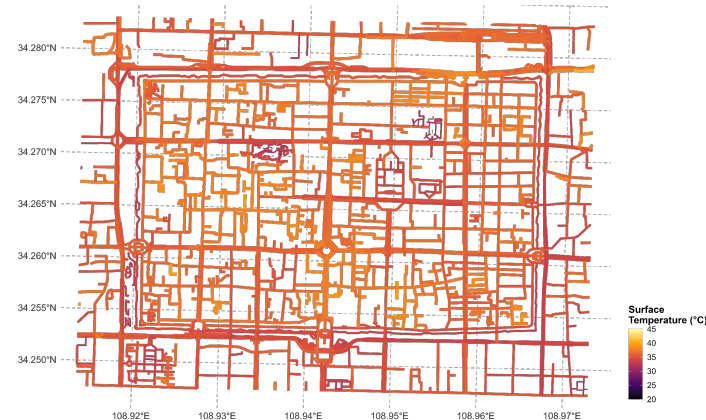
Land Surface Temperature
Computation

Method Description

Convert pixel values of the
land surface temperature
data to degrees Celsius (C)



Delft



Xian

Methodology

01 Preprocess

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Distance to
Blue/Green
Space

Method

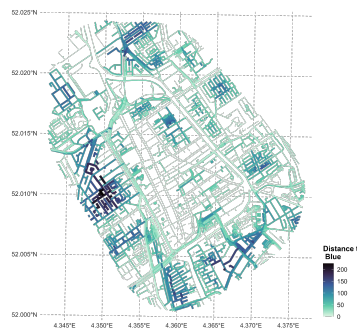
Proximity to each spaces

Method Description

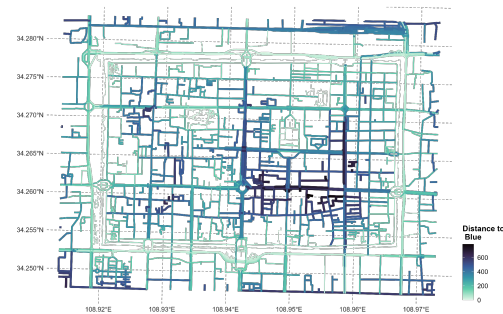
Compute the Euclidean distance to the closest green/blue space from each street segment



Distance to Green Space



Distance to Blue Space



Methodology

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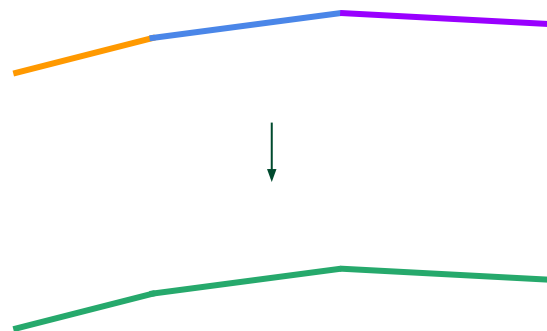
Variable Aggregation to Consolidated Streets

Method

- 1) Overlap the raw simplified street network and the consolidated network data to identify intersection
- 2) Aggregate variables for each consolidated street
 - 'Popularity': Maximum value from the overlapping segments
 - The rest: Length-weighted mean → ensure longer segments contribute proportionally more to the final values

Programming Language

R



Methodology

01 Preprocess

02 Variable Calculation

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Weighted K-Means Clustering

Purpose

Classify the street network into distinct urban typologies based on the variables

Method

- 1) Normalize (log transmission) and standardize variables
- 2) Adjust feature weights
 - Popularity: 1.0
 - LST: 2.0
 - Distance to blue/green: 0.5
- 3) Determine optimal cluster counts: Elbow method
- 4) Cluster: $K = 4$

Programming Language

R

Results

03

Results

Street Network Clusters

Cluster 1 2 3 4

Delft



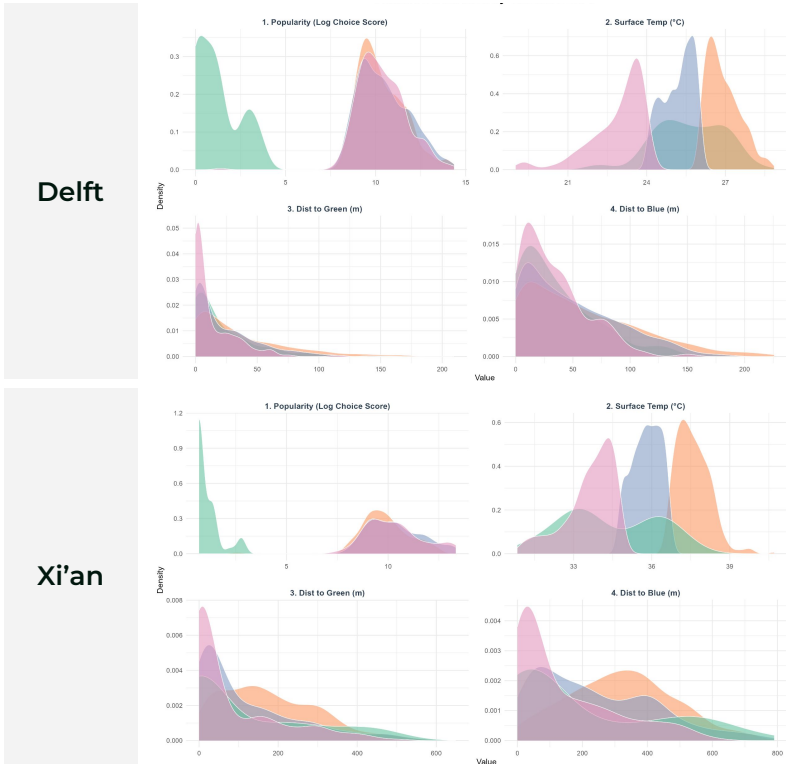
Xi'an



Results

Variable Value Range per Cluster

Cluster 1 2 3 4



Cluster	Interpretation
C1	Good thermal conditions but very low popularity
C2	High popularity despite poor thermal conditions
C3	High popularity with moderate comfort
C4	High popularity and high comfort

Results

Typology A - Underutilized Comfort Spaces

Delft

Isolated typology network overlay | City of DELFT



Oude IJsbahn



Nieuwe Gracht

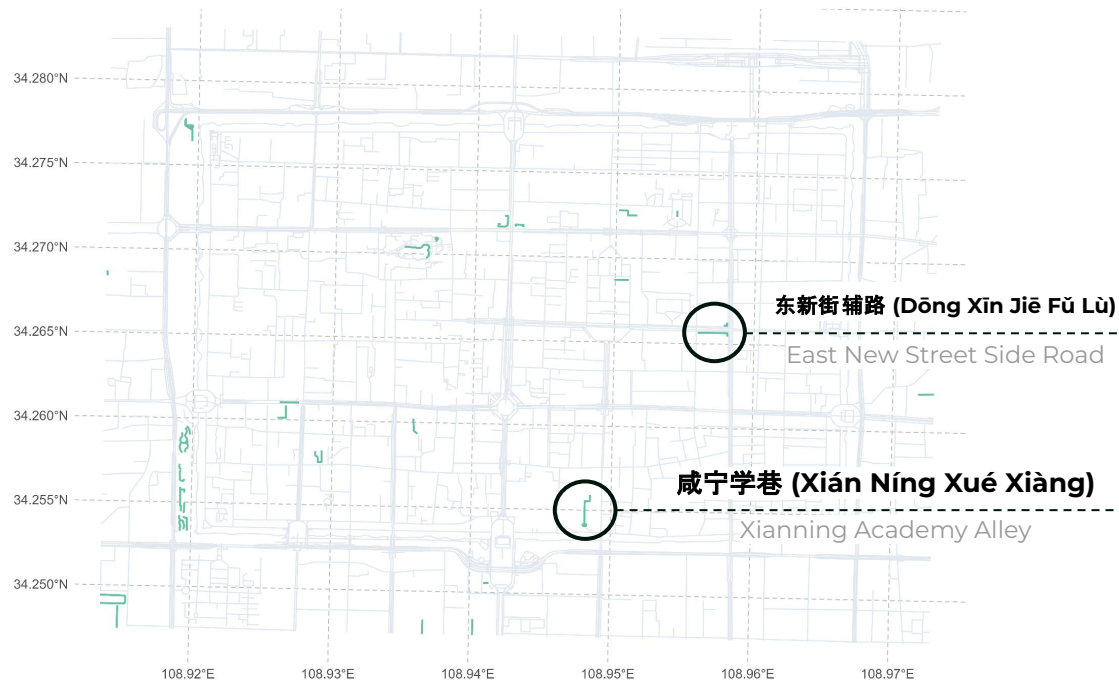


Results

Typology A - Underutilized Comfort Spaces

Xi'an

Isolated typology network overlay | City of XIAN



Results

Typology B - Function-Driven Hotspots

Delft

Isolated typology network overlay | City of DELFT



Molenstraat



Koornmarkt

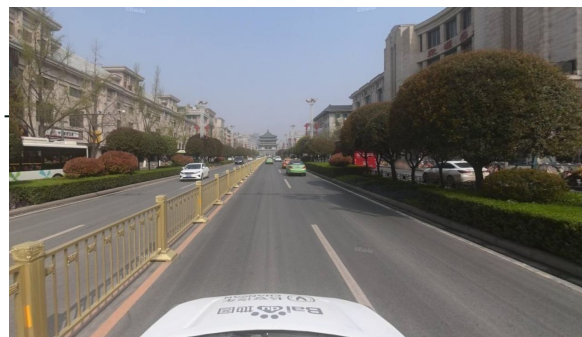
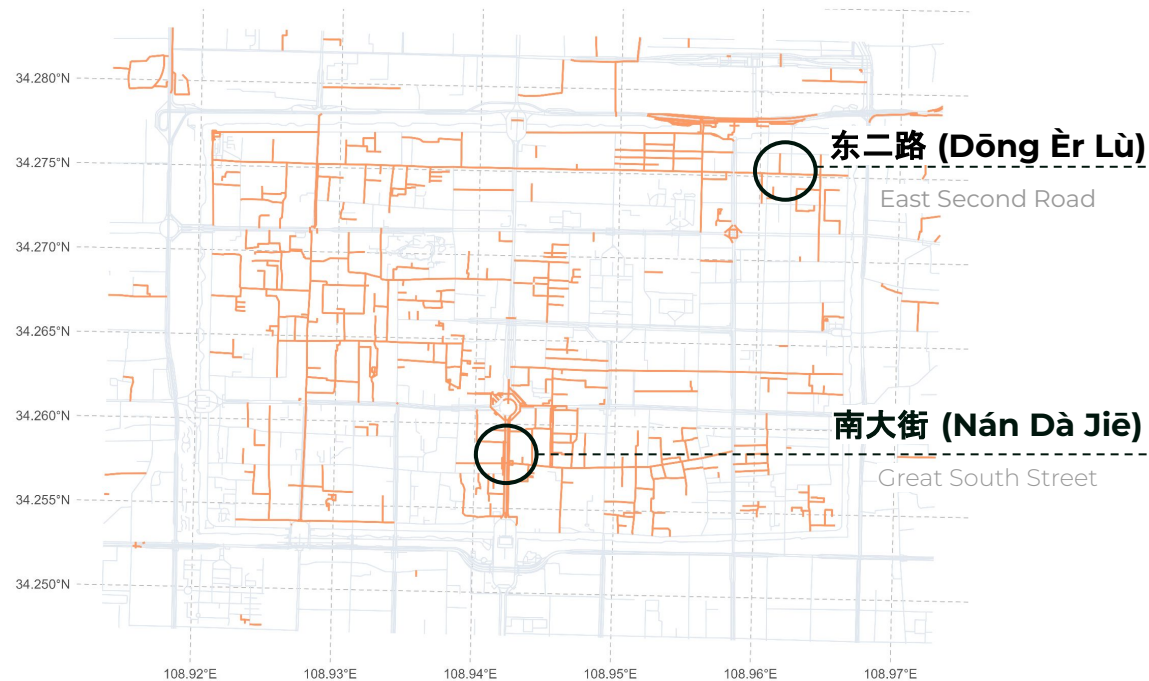


Results

Typology B - Function-Driven Hotspots

Xi'an

Isolated typology network overlay | City of XI'AN



Results

Typology C - Balanced Activity Spaces

Delft

Isolated typology network overlay | City of DELFT



Oostplantsoen



Westvest

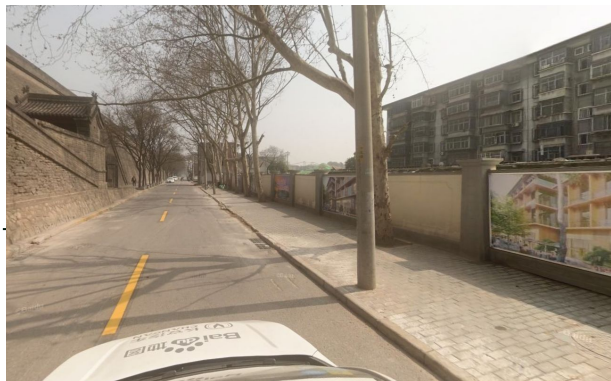
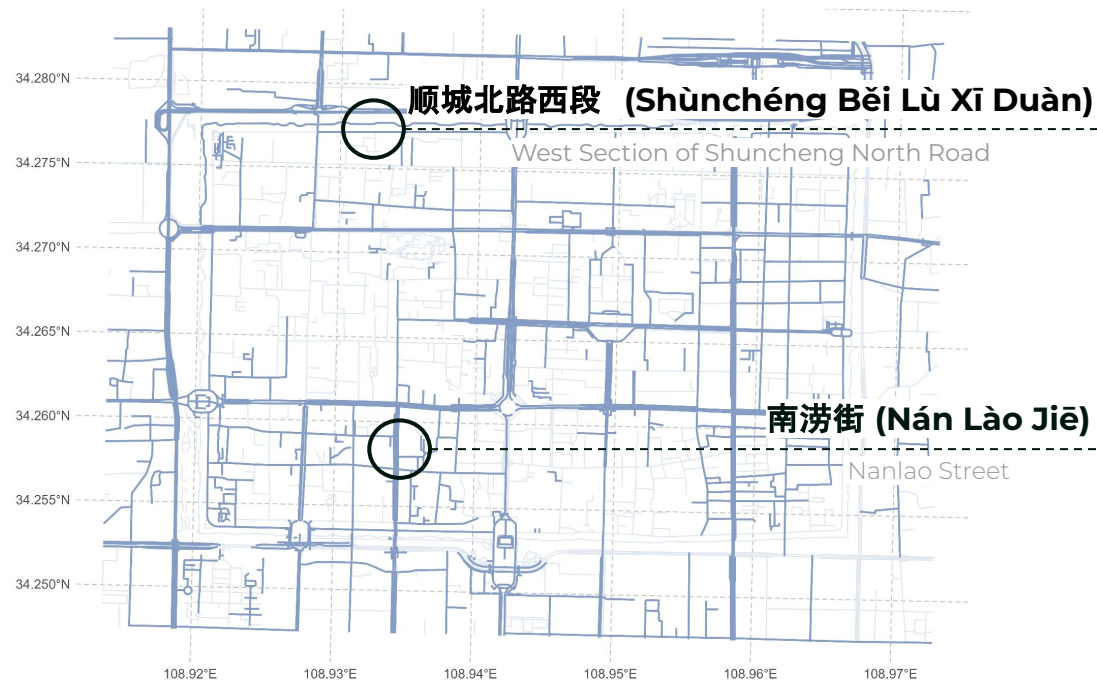


Results

Typology C - Balanced Activity Spaces

Xi'an

Isolated typology network overlay | City of XI'AN



Results

Typology D - Comfort-Driven Hotspots

Delft

Isolated typology network overlay | City of DELFT



Nieuwe Plantage



Kanaalweg

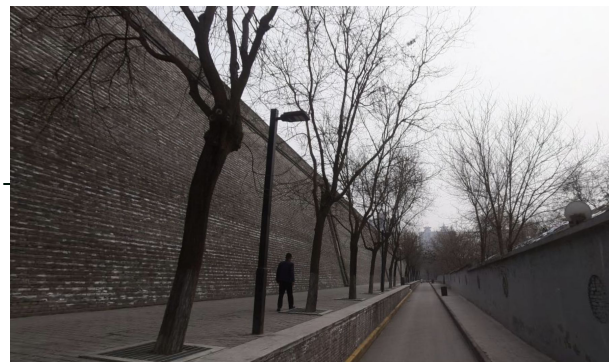
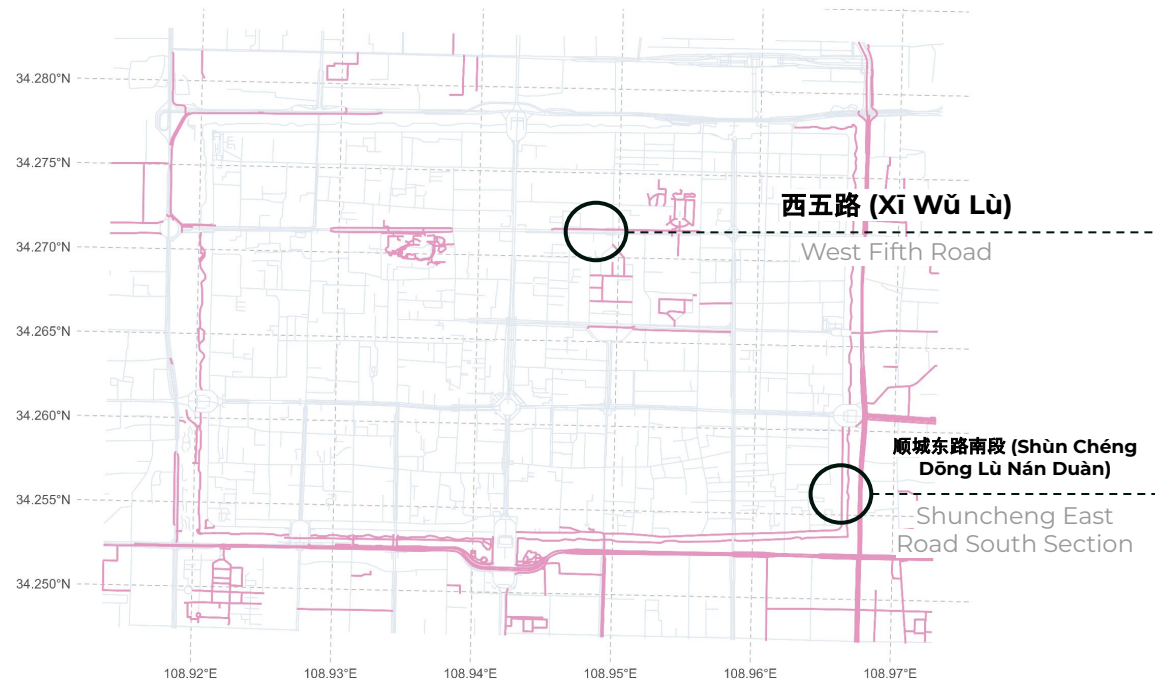


Results

Typology D - Comfort-Driven Hotspots

Xi'an

Isolated typology network overlay | City of XIAN



Discussion

04

Discussion

Pipeline Reproducibility

Standardised Data Input

Uniformly processes vector networks, satellite rasters, and environmental layers across different city scales and geometries.

Data Outputs

Automatically exports a structured suite of identical spatial layers, CSV metrics, and aligned map graphics to prevent layout shifts.

Comparative Framework

Changing a single configuration variable applies identical mathematical logic across different study areas, ensuring methodological standardization and a direct case-study comparison.

Discussion

Limitations

LST vs. Holistic Thermal Comfort

Uses **Land Surface Temperature (LST)** as a simplified proxy due to microclimate data constraints; lacks advanced indices (PET/UTCI) that incorporate humidity, wind velocity, and perceived human comfort.

Cluster Weighting

Configured **weights** (1.0 Choice, 2.0 Temperature, 0.5 Distances) to prioritize pedestrian heat exposure; these parameters can be adjusted by future researchers to align with shifting municipal goals.

Low covariance between variables

The addition of **distance variables** (to blue and green spaces) did not yield significant differences in the clustering algorithm.

Discussion

Future Development

Hierarchical Clustering

Optimize the pipeline by first grouping streets by thermal exposure and popularity, then sub-clustering those groups by distance variables to prevent dominant metrics from masking environmental buffer gaps.

Urban Interventions

Future research can be applied to focus on specific site design, managing whether a corridor requires water infrastructure or urban forestry

Mobility Management

Transportation department projects can focus on isolating high-vulnerability transit corridors for canopy expansion, shade structures, or cool-pavement coatings.

Conclusion

05

Conclusion

How does thermal comfort vary across pedestrian street segments within the historic city center?



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Thank you!

